

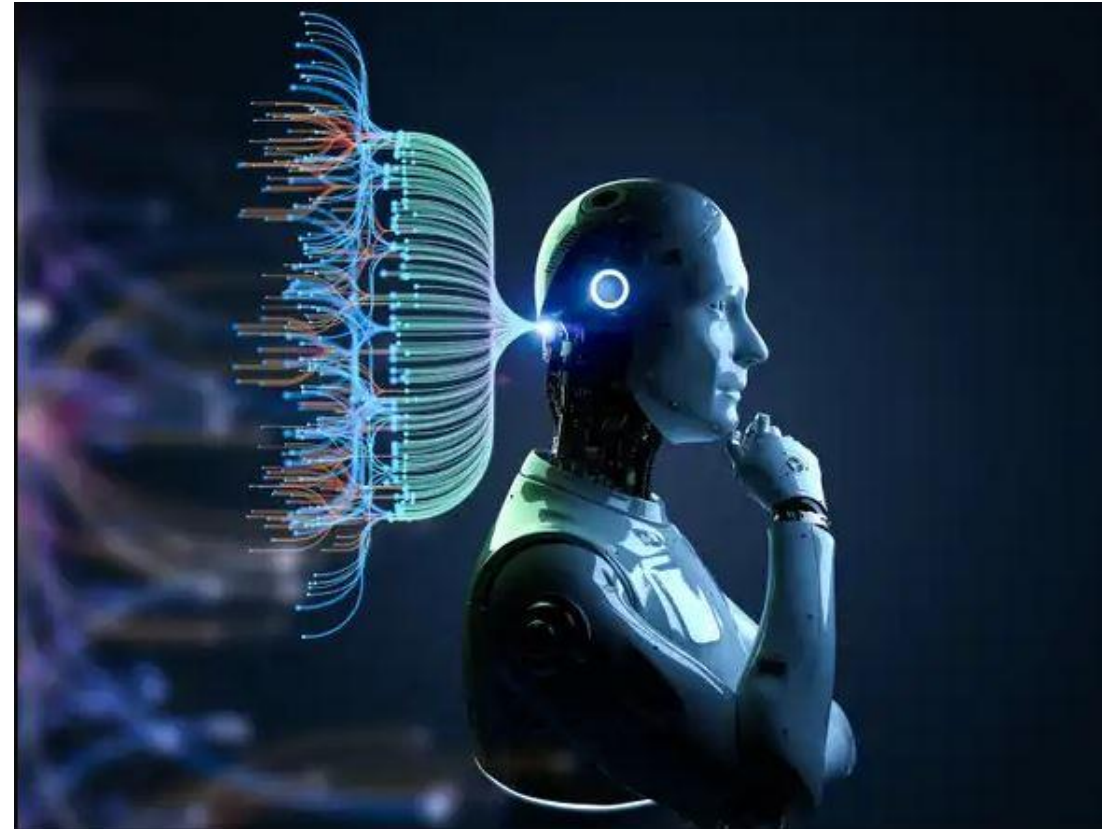
Introduction to AI

and how can it benefit our company

Agenda

Artificial Intelligence

- Definition: A logic-based technique to perform tasks that normally require human intelligence
- Biggest fields are Traditional Machine Learning and Deep Learning



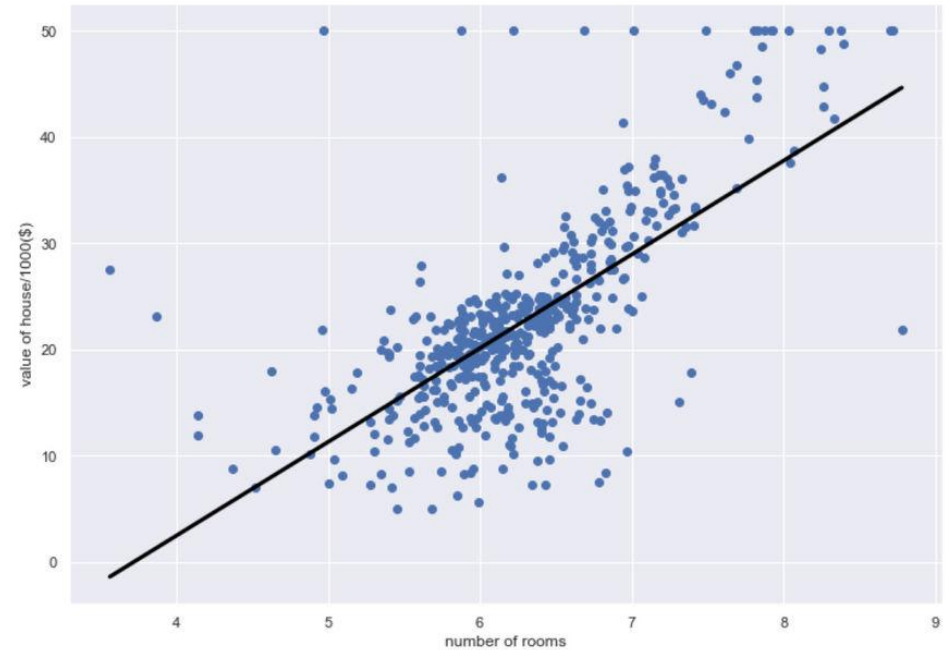
Source: <https://seekingalpha.com/article/4593012-the-softer-side-of-artificial-intelligence>

Traditional Machine Learning

- Subfield of AI
- Algorithms such as linear regression, k-nearest neighbors, and so on
- Machine Learning does not use Neural Networks
- In general faster, computationally lightweight, but can only deal with simpler tasks

Traditional Machine Learning

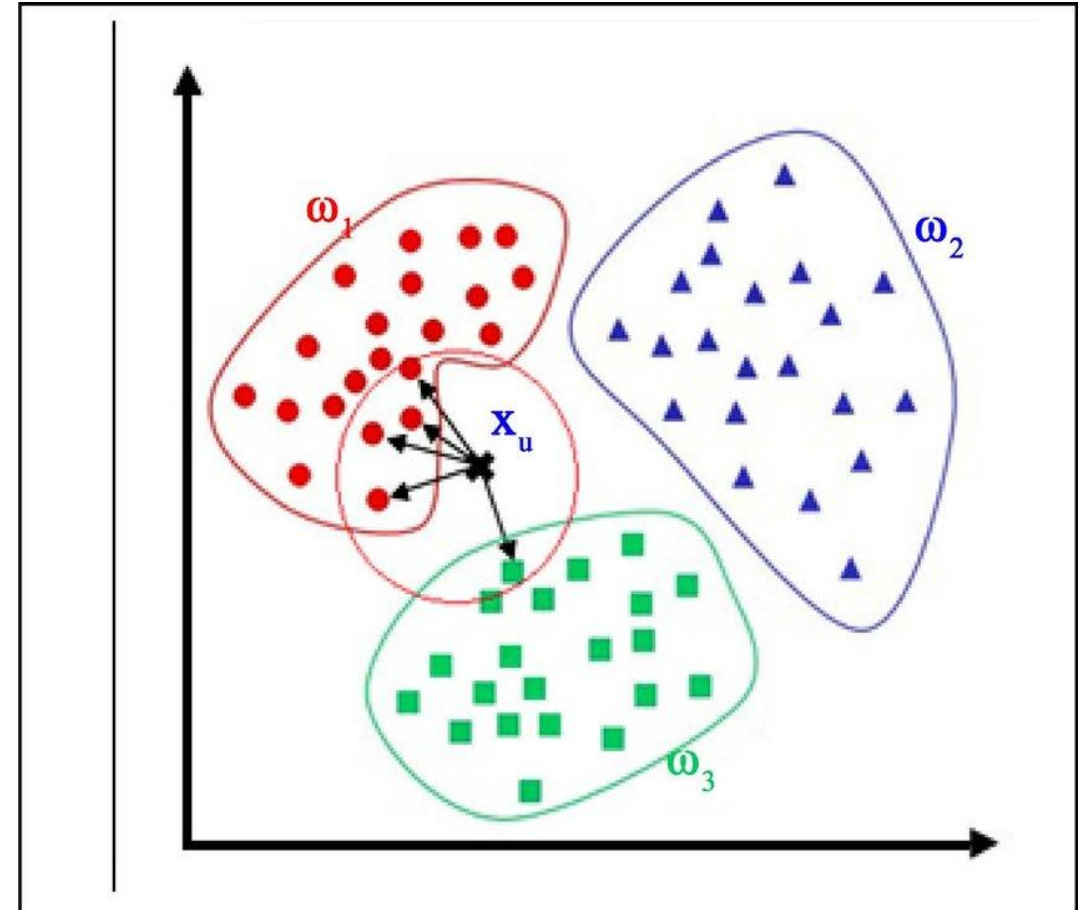
- Linear regression
- Look for a line that fits the dataset the best
- Makes prediction according to the line
- House price prediction



Source: <https://www.codersarts.com/post/predict-boston-house-prices-using-python-linear-regression>

Traditional Machine Learning

- K-nearest neighbor (k=5 in this case)
- Different colors represent different classes of data points in the dataset
- New data point is classified according to the count of nearest neighbors

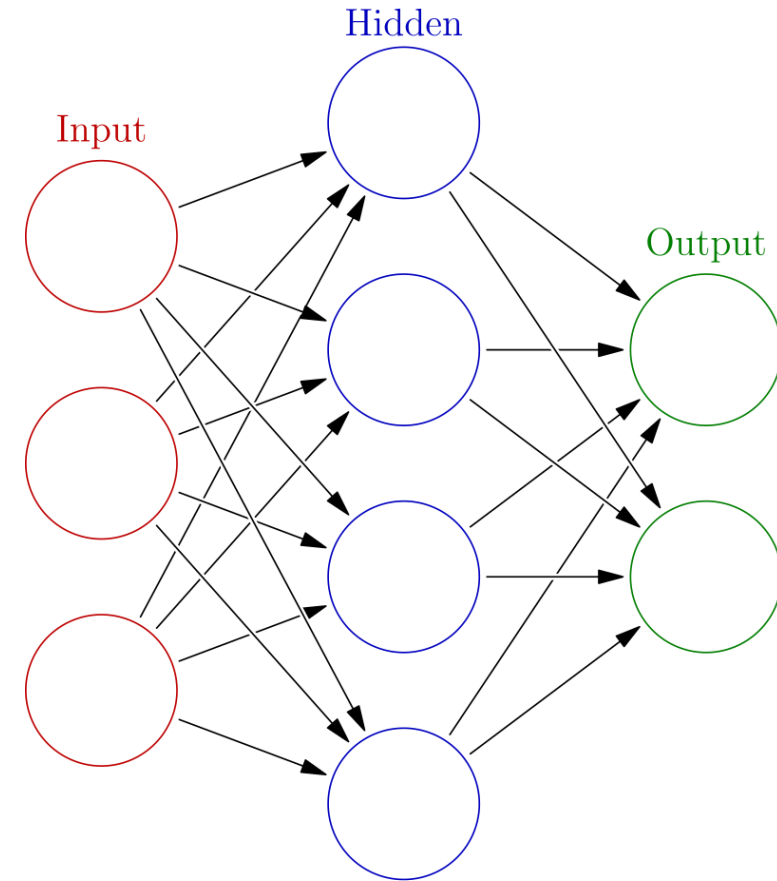


Deep Learning

- A type of machine learning that involves the use of Neural Networks
- In general computationally intensive but could deal with more difficult problems such as pattern recognition and speech recognition
- Often requires a large amount of labeled data to train the model efficiently
- Can achieve higher levels of accuracy than traditional machine learning models

Deep Learning

- Neural Network
- Input, Hidden, and Output layer
- Input layer takes Input, Hidden layer does some calculation, output received in Output layer



Deep Learning

- Forward Propagation
- Error is the difference Between the predicted outcome and the actual outcome
- Backward Propagation
- Gradient Descent to adjust the weights
- This is one epoch
- Do as many epochs until optimized result

